

Lucky Loop

Technical presentation

A predictive research lab for ML agents

Predict before compute. Verify before claim.

Qwen-AgentWorld forecasts lab actions before execution; Python produces evidence;
a deterministic verifier decides what claims survive.

Repo: github.com/enyolanev-bit/lucky-loop - source origin/main ee77b07



RESEARCH LAB OS

1. Research-lab framing

The product is not a chatbot. It is an auditable ML research loop.

0

world-model predictions accepted as scientific evidence

8

action budget in the public lab run

8

protocol seeds in the selected study

1

supported claim in current ledger

- The unit of work is a research question, not a prompt completion.
- The loop creates literature context, dataset audits, hypotheses, protocols, generated code, runs, analyses and a claim ledger.
- Every prediction is made before compute and compared with the executed outcome.
- The final report is evidence-bounded: weak or invalid claims are blocked or rewritten.

Technical thesis

Lucky Loop converts an ML research agent from a post-hoc storyteller into an instrumented lab operator: it must forecast, execute, compare, verify and preserve the audit trail.

2. End-to-end lab workflow

Question to literature to protocol to evidence to claim ledger.

Question

Natural-language research objective and compute budget

Literature

Domain and method sources, gaps, candidate hypotheses

Dataset audit

Search public datasets, reject bad candidates, select data

Protocol

Define metric, seeds, controls, claim criteria and risks

Execution

Generate, validate, dry-run and run Python experiments

Ledger

Verify claims and write supported / blocked outcomes

Lab invariant

A high score is an observation. It becomes a scientific claim only after verification.

Trace invariant

Every major state transition is written to workspace artifacts and event logs.

Reviewer invariant

The final sentence can be traced back to protocol, run, analysis and verdict.

3. Roles are deliberately separated

Planner / agent

Proposes candidate lab actions from a safe catalog. It decides what to try next.

Qwen-AgentWorld

Predicts the next lab observation, artifacts, runtime risk, failure modes and claim impact before action.

Executor

Runs real Python ML experiments, captures metrics and stores machine-readable evidence.

Comparator

Measures prediction-vs-reality, keeps misses visible and updates the lab state.

Verifier

Applies deterministic claim gates: seed noise, protocol warnings, metric discipline and claim direction.

Reporter/UI

Surfaces the complete trace and exports the report without hiding blocked claims.

4. World-model prediction contract

The model forecasts computer-lab transitions, not scientific truth.

expected observation

expected artifacts

runtime / waste risk

failure modes

protocol risk

claim impact

recommendation

Key rule Prediction is a planning signal. Only executed experiments and deterministic analysis count as evidence.

5. Safety controls before execution

Safe action catalog

Planner choices are validated against known lab actions.

Dataset audits

Public datasets are searched, audited, selected or rejected with rationale.

Protocol criteria

The protocol fixes primary metric, seeds, controls and claim thresholds.

Static validation

Generated experiment code is checked before full execution.

Dry run

Plumbing failures are caught before longer runs.

Protocol warnings

Leakage, split invalidity or metric misuse can block a claim even if the score is high.

6. Verifier and claim ledger

Scientific output is a verdict with evidence IDs.

Current deterministic gate

```
effect_size = best_mean_metric - worst_mean_metric
seed_noise = max(best_config_seeds) - min(best_config_seeds)
effect_to_noise_ratio = effect_size / seed_noise
protocol_warning => claim blocked
```

0.346

ledger effect_size

0.022

seed_noise

42.9

ratio

Current ledger example

Claim: A Transformer-based model (e.g., HART) robustly outperforms logistic regression on public sensor classification data across repeated seeds. Verdict: supported. Reason: The measured effect exceeded seed noise by the support threshold.

Integrity note

origin/main also contains an older contradictory study_result entry for this workspace. Before a formal demo, harmonize study_result, final_report and claim_ledger.

7. Public lab run snapshot

A concrete workspace produced by the research lab.

8

action budget

8

repeated seeds

eeg_eye_state

selected dataset

balanced_accuracy

primary metric

Research question

Can a nonlinear model robustly outperform logistic regression on public sensor classification data across repeated seeds?

Conditions tested

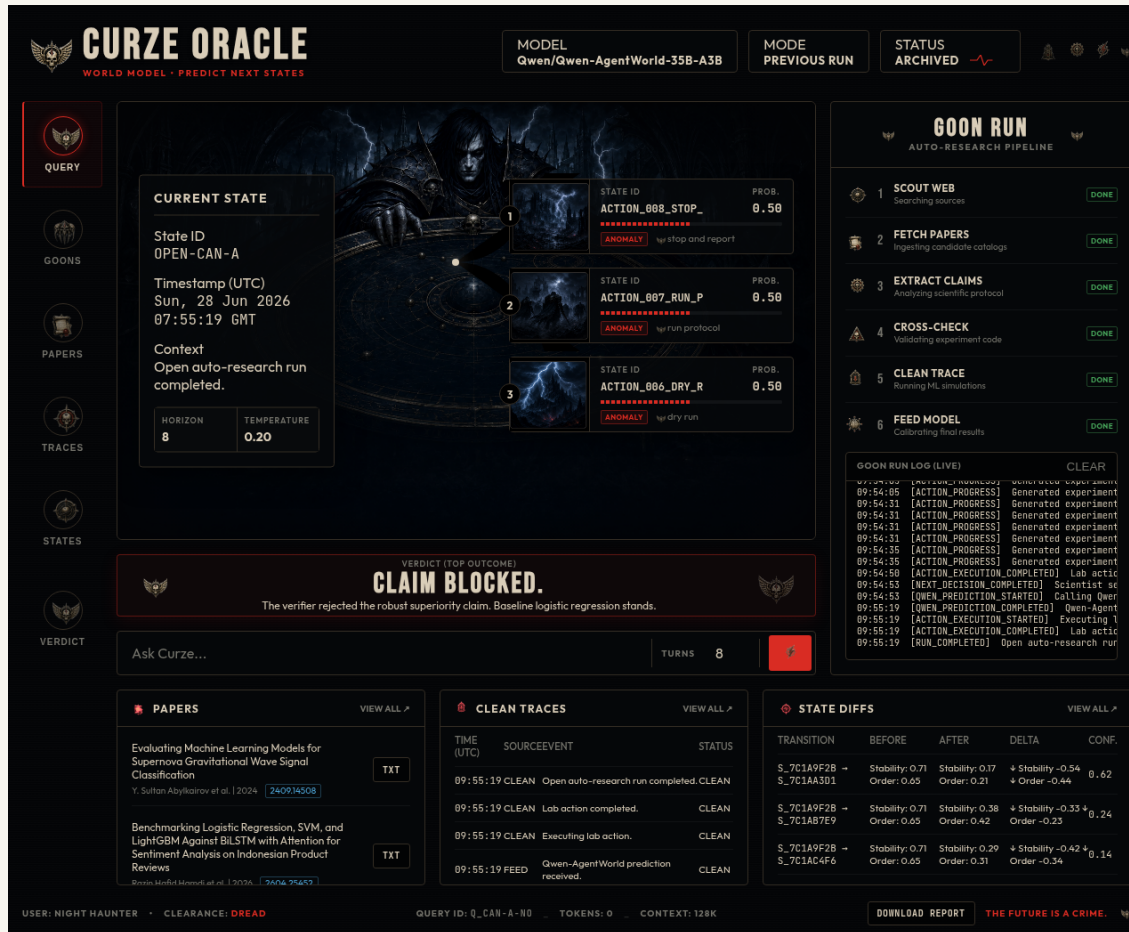
gradientboostingclassifier, logisticregression, mlpclassifier, randomforestclassifier, svc

Protocol risks

generated_protocol, dataset_selection_bias

The UI is an audit cockpit, not just a result viewer.

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Frontend

Static HTML/CSS/JS served by web/server.py. No Node build step.

- GET /api/config
- GET /api/previous-runs
- POST /api/run
- GET /api/status?slug=...
- GET /api/report?slug=...

What the reviewer sees

Current state, predicted branches, live events, papers, clean traces, state diffs, verdict banner and downloadable report.

9. Operating the lab

Minimal commands for a technical demo.

```
python3 -m venv .venv
. .venv/bin/activate
python -m pip install -r requirements.txt
set -a; source .env; set +a
PYTHONPATH=src python web/server.py 8000

PYTHONPATH=src python -m luckyloop.lab \
  --question "Can a nonlinear model robustly outperform logistic regression..." \
  --budget 8
```

LLM config

Planner and world model are separate OpenAI-compatible endpoints.

Smoke mode

A no-model mode exists to test local plumbing without credentials.

Validation

scripts/validate_lab_artifacts.py checks artifact presence and minimal coherence.

10. Reviewer risks and honest boundaries

Statistics

The effect/noise gate is useful but not a full significance engine. Add paired tests or bootstrap intervals for strong claims.

Benchmark scope

Current evidence is mostly sklearn/tabular and one public lab run. Do not claim general scientific discovery.

World-model calibration

Qwen-AgentWorld is a planning signal; its misses must remain in the report.

Artifact drift

The current repo has one consistency issue between study_result and claim_ledger.

Dataset selection bias

Automatic dataset search can bias the study. Keep rejected candidates and audits visible.

Generated code

Static validation reduces risk but does not prove scientific correctness.

Final technical message

Lucky Loop is a research lab loop where the agent must forecast before it acts.

The world model guides compute allocation, but real Python execution creates the evidence.

The verifier and claim ledger turn the final report into an auditable scientific object.

PREDICT BEFORE COMPUTE

VERIFY BEFORE CLAIM

AUDITABLE RESEARCH LAB